

TANISHKA

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SUMMARY

B.Tech Computer Science student specializing in Machine Learning and Natural Language Processing with hands-on experience in building end-to-end text classification systems. Proficient in Python, TensorFlow, and Scikit-learn with experience in deep learning models (LSTM, CNN-LSTM) and feature engineering techniques (TF-IDF, Word2Vec, FastText). Experienced in building and deploying ML models for real-world applications.

SKILLS

- Technical Skills: Machine Learning, Deep Learning, NLP
 - Programming Languages: Python , Java , C , C++ , R
 - Libraries & Tools: TensorFlow, Scikit-learn, Pandas, NumPy, NLTK, Matplotlib, Seaborn, Jupyter Notebook
 - Databases: MySQL, MongoDB
 - Core Subjects: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks
 - Soft Skills - Communication , Problem Solving , Teamwork , Project Management , Public Speaking
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WORK EXPERIENCE

AI Intern

Amity School of Engineering & Technology | June 2025 – July 2025

- Built an end-to-end machine learning pipeline for automated text classification
 - Performed data preprocessing including tokenization, stemming, stopwords removal, and text normalization
 - Applied feature engineering techniques such as TF-IDF and feature selection methods
 - Implemented deep learning models including LSTM and improved performance using hyperparameter tuning and regularization
 - Achieved 97% model accuracy using LSTM architecture
 - Evaluated models using accuracy, precision, recall, F1-score, and confusion matrix
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PROJECTS

Cyberbullying Detection Model

- Designed a multi-class text classification system for detecting harmful social media content
- Implemented ML and DL models for benchmarking performance
- Generated embeddings using Word2Vec, FastText, and GloVe with PCA
- Performed sentiment analysis and feature engineering on large-scale Twitter datasets
- Compared multiple models to identify the best-performing architecture

Speech Emotion Recognition System (ML & DL)

- Developed an end-to-end SER system using the RAVDESS dataset for multi-class emotion classification
- Extracted audio features (MFCC, Chroma, Mel Spectrogram) using Librosa
- Built and compared ML (SVM, Random Forest, LightGBM) and DL models (CNN, LSTM, CNN-BiLSTM)
- Achieved 90% accuracy using CNN-BiLSTM, capturing spatial and temporal patterns
- Developed a web interface (HTML, CSS) with backend integration using Flask (app.py)

Fake News Detection System

- Developed NLP pipeline using NLTK (tokenization, stemming, stopwords removal)
 - Applied feature extraction (TF-IDF, PCA, Truncated SVD)
 - Built models: Logistic Regression, SVM, LSTM, CNN-LSTM, CNN-BiLSTM
 - Improved performance using hyperparameter tuning and class balancing
 - Evaluated using accuracy, precision, recall, F1-score
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EDUCATION

B.Tech – Computer Science & Engineering

Amity University Punjab | Aug 2023 – Present

CGPA: 7.86 / 10

Senior Secondary (Non-Medical)

GMSSS 23A Chandigarh | 2021 – 2023

CERTIFICATIONS

[Complete Data Science, Machine Learning, DL, NLP Bootcamp By Krish Naik](#)